

Learning under Partial Knowledge:

Robust Generalisation and Imprecise Forecasting

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The Dogmatism of Modern Machine Learning

Tradeoff

$$\mathscr{L}(\theta) = \mathscr{L}_{data}(\theta) + \lambda\Omega(\|\theta\|_{\Theta}) + \varepsilon, \quad \theta \in \Theta$$

Data
what we *already* know

- loss/utility functions
- likelihood functions
- score functions

Inductive Bias
what we *kind of* know

- neural network architectures
- prior knowledge
- symmetry and invariance

Anything Else
what we *don't* know

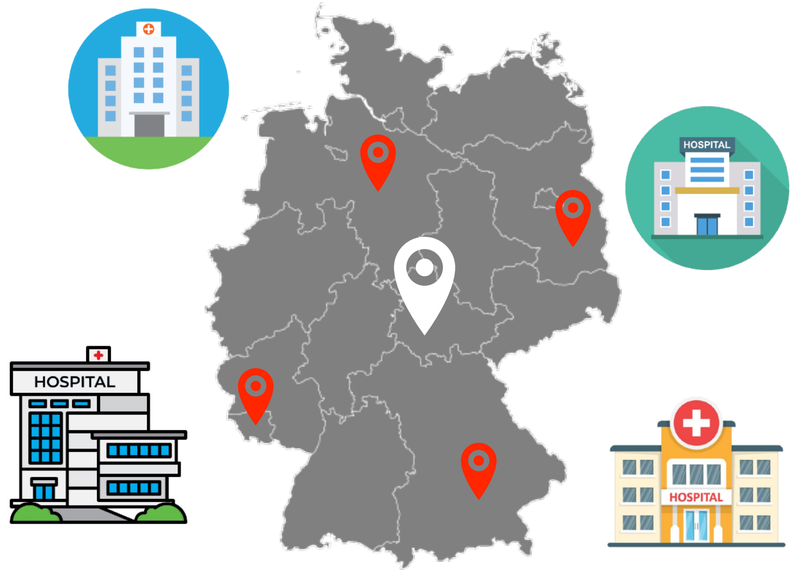
- heterogeneity
- preferences
- normative judgement
- unknown unknowns

Can we turn this blind spot into an object of inference?

This talk: Domain generalisation and imprecise forecasting

Domain Generalisation (DG)

- A collection of training distributions: $P_1(X, Y),\ P_2(X, Y),\ \ldots,\ P_N(X, Y) \rightarrow P^*$
  - How to take knowledge acquired from an arbitrary number of related domains and apply it to previously **unseen** domains?
- Popular approaches:
  - **Invariance principle** (Muandet et al. 2013; Peters et al., 2016; Arjovsky et al., 2019)
  - **Meta learning** (Li et al., 2017)
  - **DRO** (BenTal et al., 2009; Sagawa et al., 2020)
  - **Risk extrapolation** (Krueger et al., 2021)



Blanchard et al. NeurIPS (2011); Muandet et al. ICML (2013)

Dogmatism

the tendency to hold beliefs with unwarranted certainty

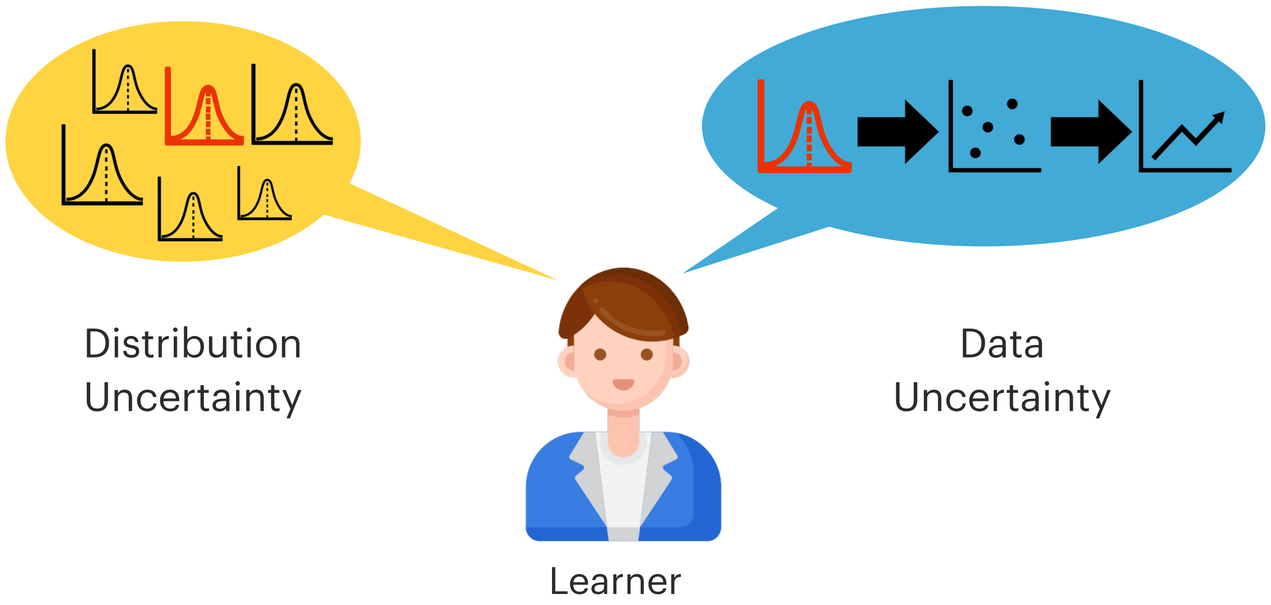
Dogmatic Generalisation

Precise (or dogmatic) learner deals with two sources of uncertainties simultaneously.

1. The learner chooses **how to generalise** (pick a specific distribution)
2. The learner then **learns the best hypothesis** (statistical learning)

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_{(X,Y)\sim P_i}[\ell(f(X),Y)]$$

$$\max_{P\in\{P_1,\ldots,P_N\}}\mathbb{E}_{(X,Y)\sim P}[\ell(f(X),Y)]$$

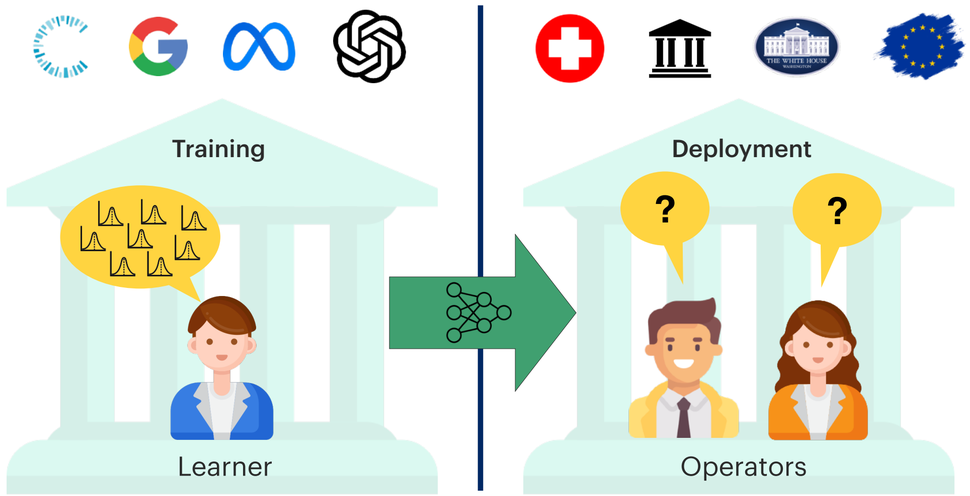


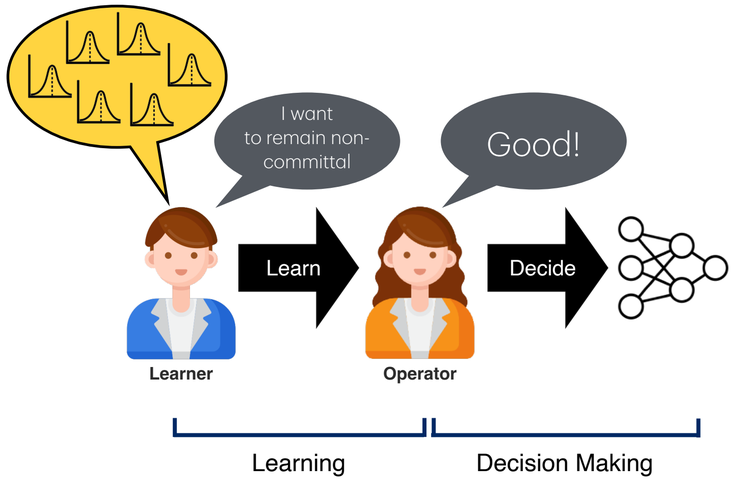
$$\max_{P\in\{\hat{P}_1,\ldots,\hat{P}_N\}}\frac{1}{m}\sum_{j=1}^{m}\ell(f(x_i),y_i)$$

Institutional Separation

Precise (or dogmatic) learner deals with two sources of uncertainties simultaneously.

1. **The learner chooses how to generalise (pick a specific distribution)**
2. The learner then learns the best hypothesis (statistical learning)





Manski, Charles F. Statistical Inference for Statistical Decisions. 2019

Domain Generalisation via Imprecise Learning



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ICML International Conference On Machine Learning

Spotlight

Domain Generalisation via Imprecise Learning

Anurag Singh [1] **Siu Lun Chau** [1] **Shahine Bouabid** [2] **Krikamol Muandet** [1]

Abstract

Out-of-distribution (OOD) generalisation is challenging because it involves not only learning from empirical data, but also deciding among various notions of generalisation, e.g., optimising the average-case risk, worst-case risk, or interpolations thereof. While this choice should in principle be informed by the relevant deployment scenarios, in practice it is often made implicitly by default as the chosen learning algorithm encodes certain assumptions about the data-generating process. When these assumptions fail, models can exhibit unexpectedly poor and brittle generalisation performance. In this paper, we propose to explicitly account for this ambiguity by learning sets of predictive models that are compatible with the observed data. To this end, we propose a novel OOD generalisation algorithm based on the theory of imprecise probabilities. Our method learns an outer approximation of the risk and its corresponding minimiser, which can be used to obtain guarantees against worst-case OOD risk and to identify predictions that are prone to unreliable generalisation. We demonstrate the practical utility of our approach on synthetic and real-world datasets, showing that it can reliably detect situations where standard OOD generalisation assumptions break down.

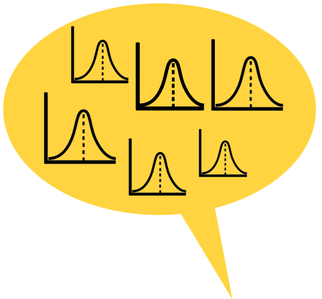
1. Introduction

Machine learning models have achieved remarkable success in recent years, with some systems (LLM) that surpass human-level generalisation capabilities in specific domains.

Despite notable achievements, these systems may catastrophically fail when operated on out-of-domain (OOD) data because theoretical guarantees for their generalisation hinge on the assumption of independent and identically distributed (IID) training and deployment data, with empirical

Problem Formulation

Credal Set





Imprecise Learner

- A **risk profile** on n observed environments $P_1(X, Y), P_2(X, Y), \ldots, P_N(X, Y)$

$$\mathbf{R}(f) := (R_1(f), \ldots, R_N(f)), \quad f \in H$$

- An **aggregation function** $\rho_\lambda : L_2^N(H) \to L_2(H)$ for some $\lambda \in \Lambda$

- For a fixed $\lambda \in \Lambda$, we can learn from H by minimising an aggregated risk

$$f_\lambda^* = \arg\min_{f \in H} \rho_\lambda[\mathbf{R}](f), \quad \lambda \in \Lambda$$

- Learn an **augmented hypothesis** $h_\theta : H \times \Lambda \to \mathcal{Y}$ such that

Traverse credal set

$$h_\theta^*(\cdot, \lambda) = f_\lambda^* = \arg\min_{f \in H} \rho_\lambda[\mathbf{R}](f), \quad \lambda \in \Lambda$$

Problem Formulation

Aggregation function

$$\rho_\lambda : L_2^N(H) \to L_2(H)$$

Risk profile on n observed environments

$$\mathbf{R} := (R_1(f), \ldots, R_N(f)), \quad f \in H$$

$$h^* = \arg\min_{\theta \in \Theta} \rho_\lambda[\mathbf{R}]\big(h_\theta(\,\cdot\,, \lambda)\big), \quad \lambda \in \Lambda$$

Parameter Space



Augmented Hypothesis

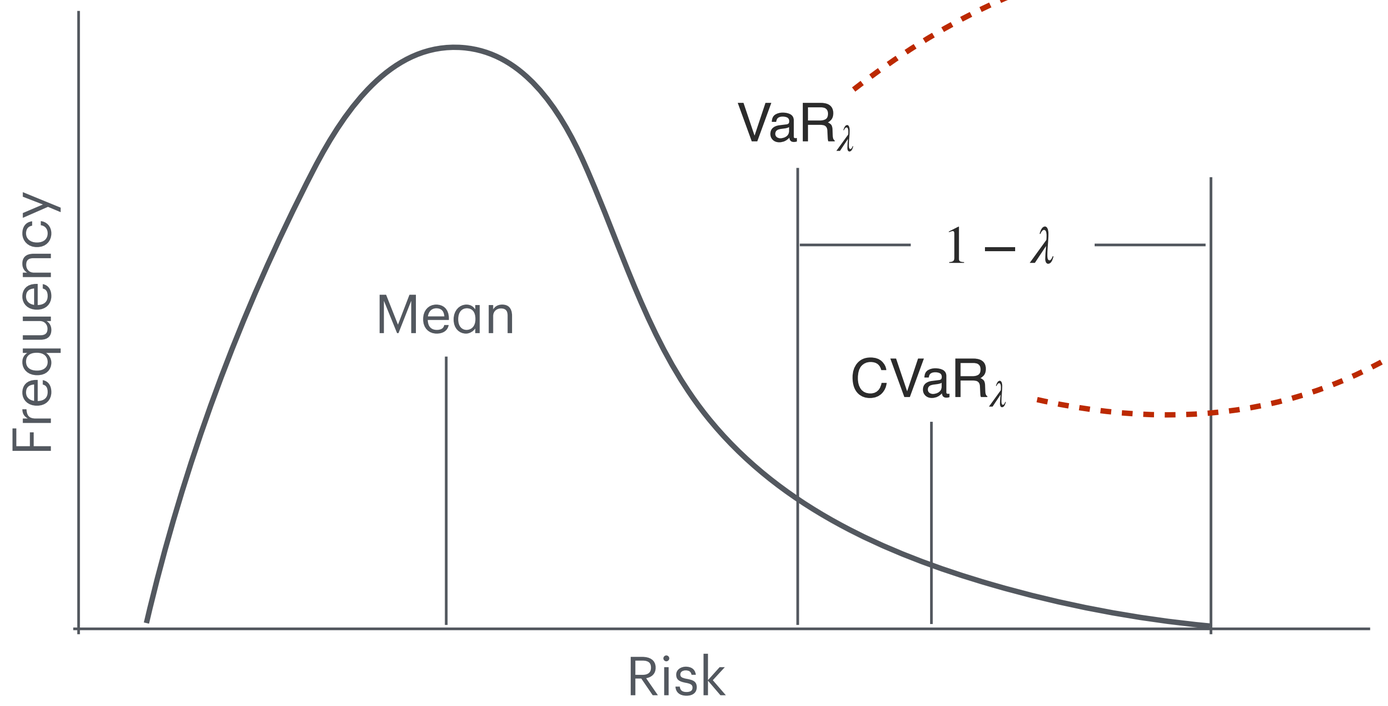
$$h \in H \times \Lambda$$

$$h(\,\cdot\,, \lambda) \in H, \ \ \lambda \in \Lambda$$

Choice Space

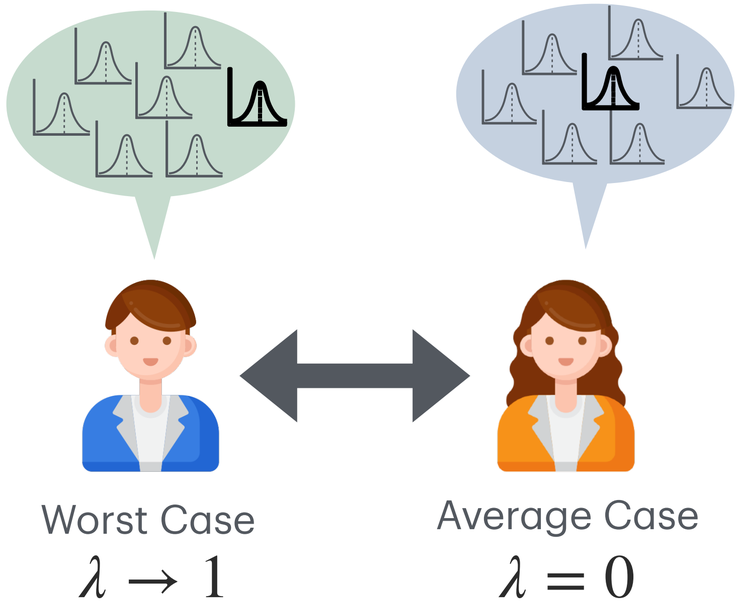


Conditional Value at Risk (CVaR)



$$\text{VaR}_\lambda(R) = \min\{r \mid F_R(r) \geq \lambda\}$$

$$\text{CVaR}_\lambda(R) = \mathbb{E}\left[R \mid R > \text{VaR}_\lambda(R)\right]$$



- **Interpretation**: λ is the level of risk aversion
  (Robey et al., 2022; Eastwood et al., 2022a; Li et al., 2023)

C-Pareto Optimality

Dominance

h_θ dominates h'_θ if for all $\lambda \in \Lambda$,

$$\rho_\lambda[\mathbf{R}](h_\theta(\,\cdot\,,\lambda)) \leq \rho_\lambda[\mathbf{R}](h'_\theta(\,\cdot\,,\lambda))$$



Scalarised Objective

For $Q \in \Delta(\Lambda)$ with **full support**,

$$J_Q(h_\theta) := \mathbb{E}_{\lambda\sim Q}\left[\rho_\lambda[\mathbf{R}](h_\theta(\,\cdot\,,\lambda))\right]$$

- Does the specific choice of Q matter?
- Which Q should we choose to represent our ignorance about λ? *A uniform distribution?*
- Must we commit to a single Q, can we remain imprecise?

Imprecise Risk Optimisation (IRO)

Algorithm 1 Imprecise Risk Optimisation (IRO)

Input: Data from d distinct domains $\{x_i^{(d)}, y_i^{(d)}\}_{i=1}^n \sim \mathbb{P}_d(X, Y)$, a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$, a probability space $\Delta(\Lambda)$, a (augmented) hypothesis class $\mathcal{H}_\Lambda$, risk aggregator $\rho_\lambda : \mathcal{L}^d(\mathcal{H}) \to \mathcal{L}(\mathcal{H})$, number of Monte Carlo samples m.
Initialise the parameter $\xi \in \Xi$.
repeat
  Estimate Q_t^* with $\widehat{Q}_t$ by solving (9) by computing $\widehat{Q}_t = \arg\min_{Q \in \Delta(\Lambda)} \| \frac{1}{m} \sum_{j=1}^m \nabla \rho_{\lambda_j} [\widehat{\boldsymbol{\mathcal{R}}}](\bar{h}_\xi(\cdot, \lambda_j)) \|_2$ where $\lambda_1, \ldots, \lambda_m \sim Q$.
  Compute $\hat{v}_t(\xi) = \frac{1}{m'} \sum_{k=1}^{m'} \nabla \rho_{\lambda_k} [\widehat{\boldsymbol{\mathcal{R}}}](\bar{h}_\xi(\cdot, \lambda_k))$ where $\lambda_1, \ldots, \lambda_{m'} \sim \widehat{Q}_t$.
  Update $\xi = \xi - \eta \hat{v}_t(\xi)$.
until $\|\hat{v}_t(\xi)\|_2 > \epsilon$

- Is **C-Pareto improvement** still possible?

$$Q_t \in \arg \min_{Q \in \Delta(\Lambda)} \left\| \nabla_{\theta_{t-1}} \hat{J}_Q \left(h_{\theta_{t-1}} \right) \right\|_2$$

- If yes, we update the parameters θ_t

$$\theta_t \leftarrow \theta_{t-1} - \eta \nabla_\theta \hat{J}_{Q_t}(h_\theta)$$

$$\hat{J}_{Q_t}(h_\theta) := \frac{1}{m} \sum_{i=1}^m \rho_{\lambda_i}[R](h_\theta(\cdot, \lambda_i))$$

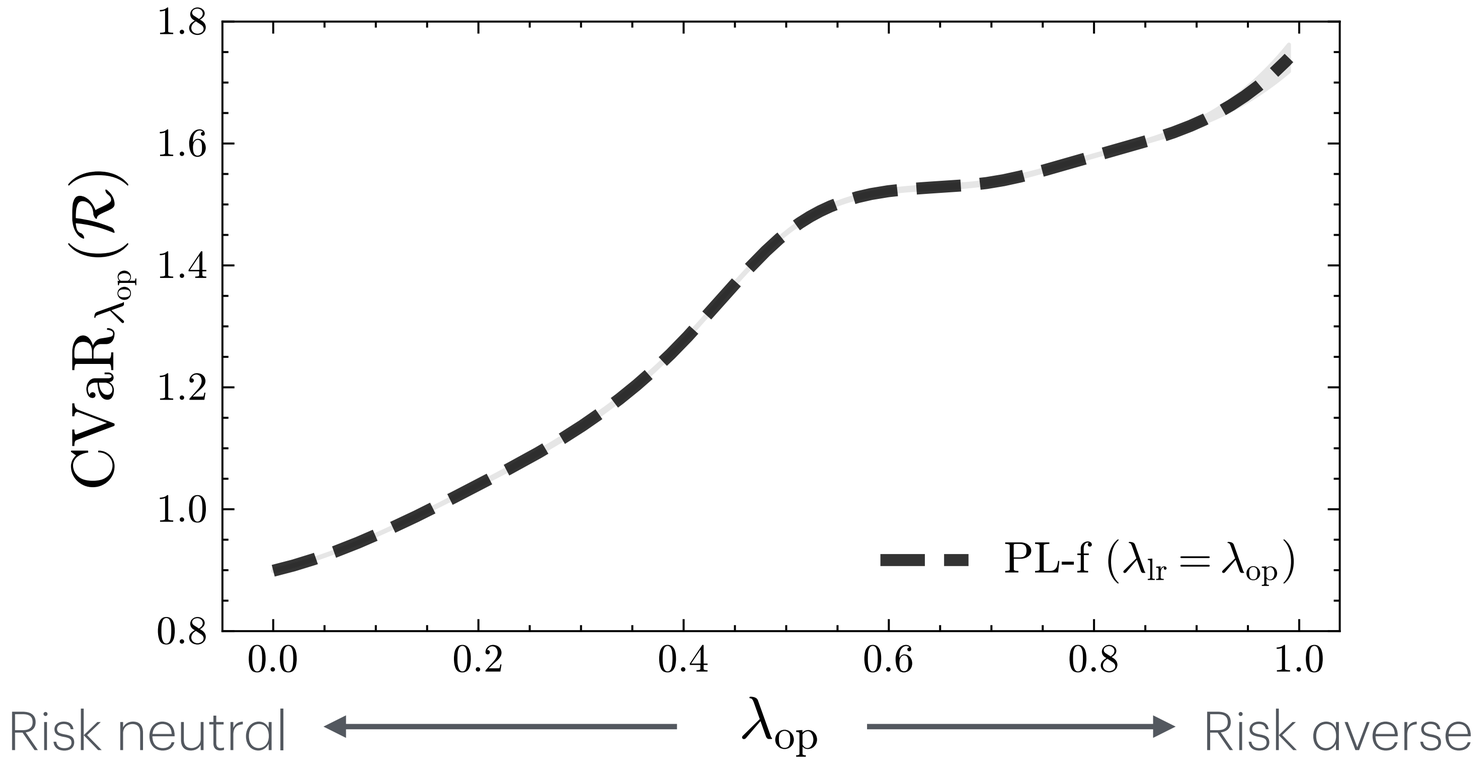
$$\lambda_1, \ldots, \lambda_m \sim Q_t$$

- Similar to the **multiple-gradient descent algorithm (MGDA)** (Desideri, 2012).

Precise vs Imprecise Learning

$$Y_d = \theta_d X + \varepsilon,\ X \sim \mathcal{N}(1,0.5),\ \epsilon \sim \mathcal{N}(0,0.1),\ \theta_d \sim \mathcal{U}(1,1.1) \text{ or } \mathcal{U}(-1.1,-1)$$

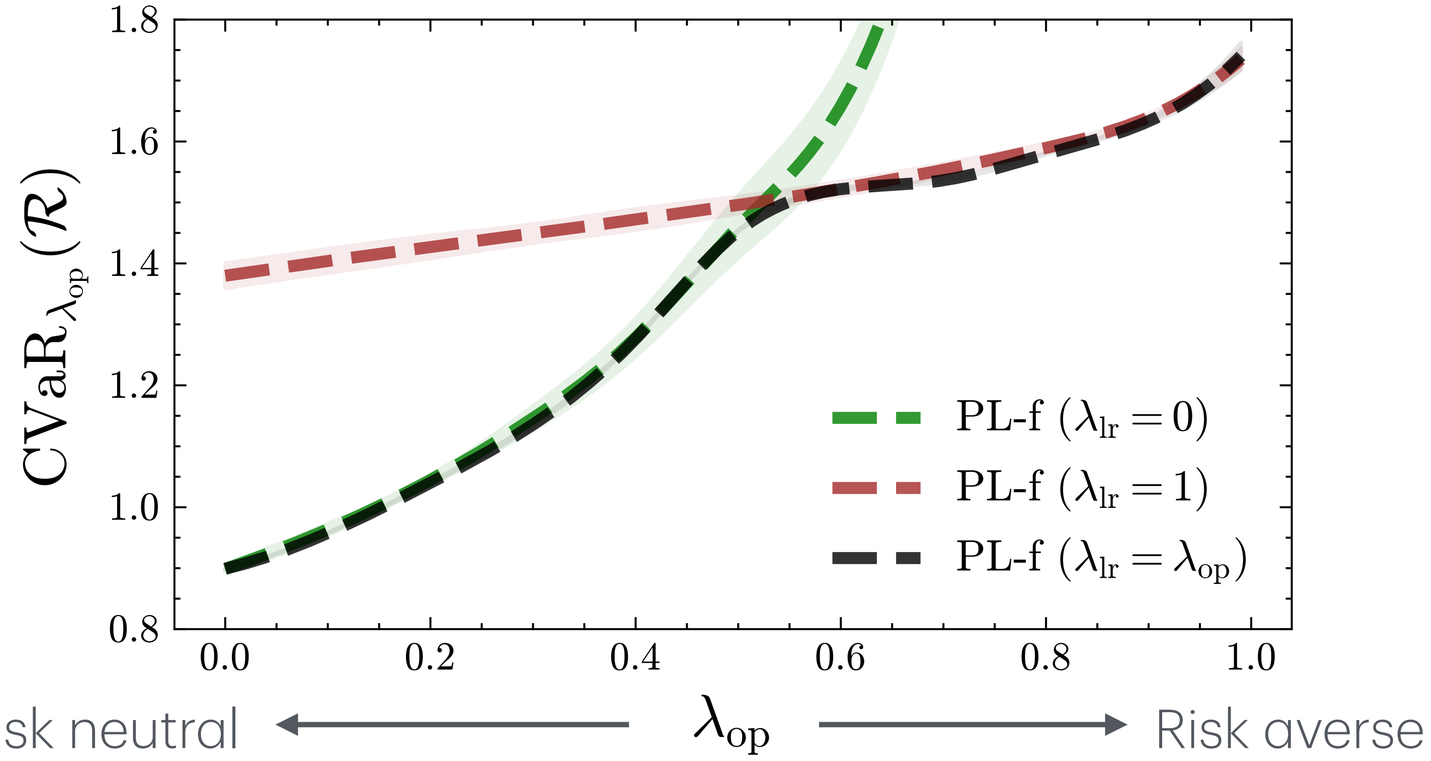
n_{train} = 250, n_{test} = 250, sample size = 100 from each domain



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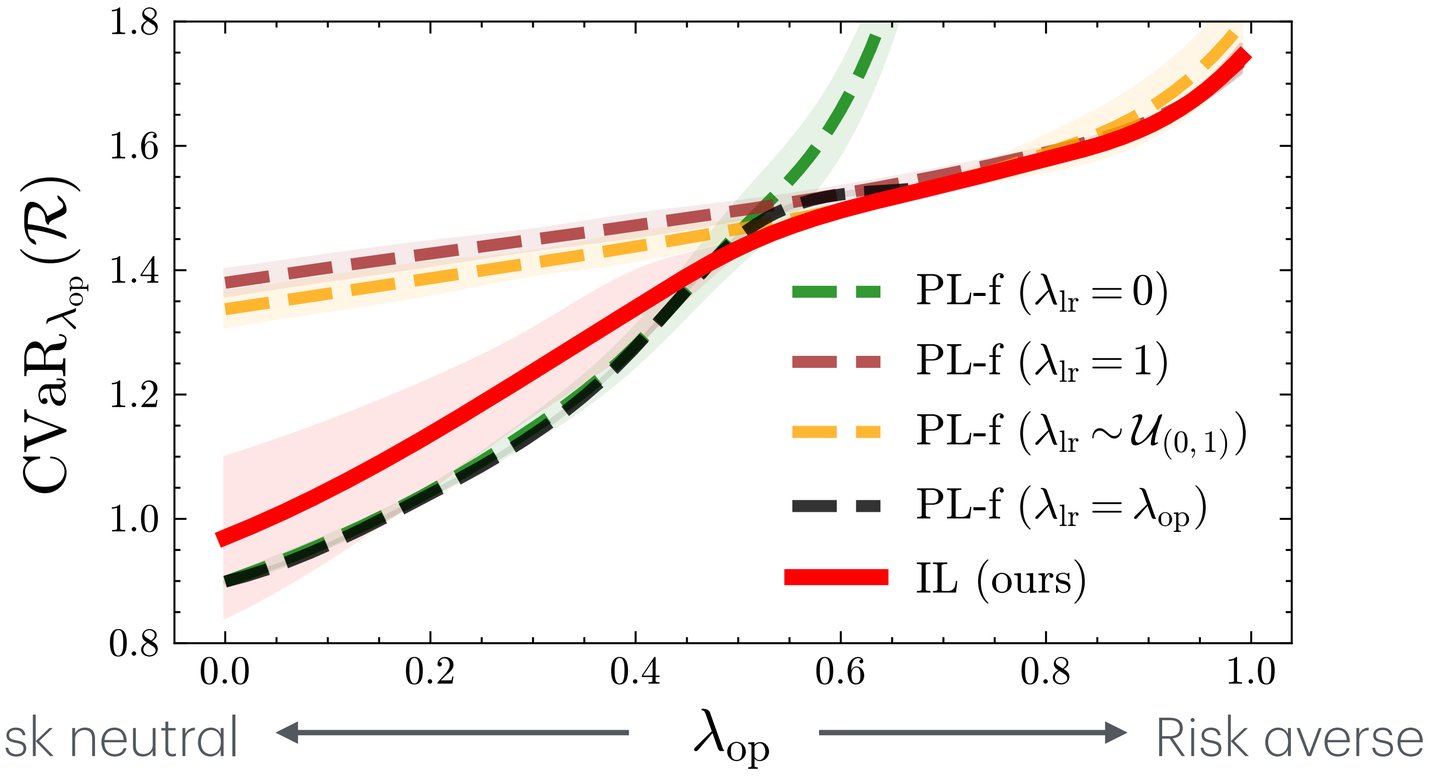
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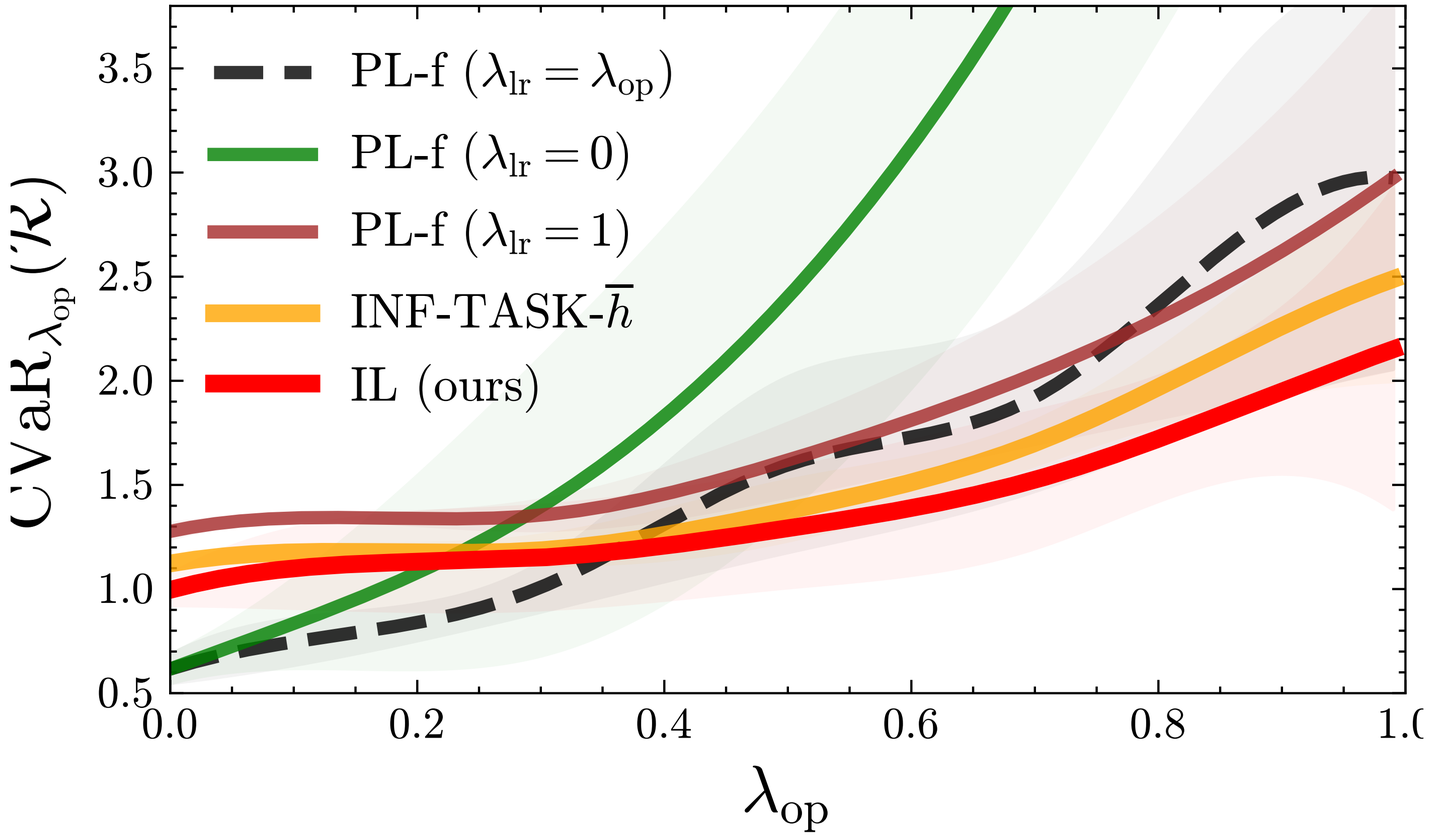
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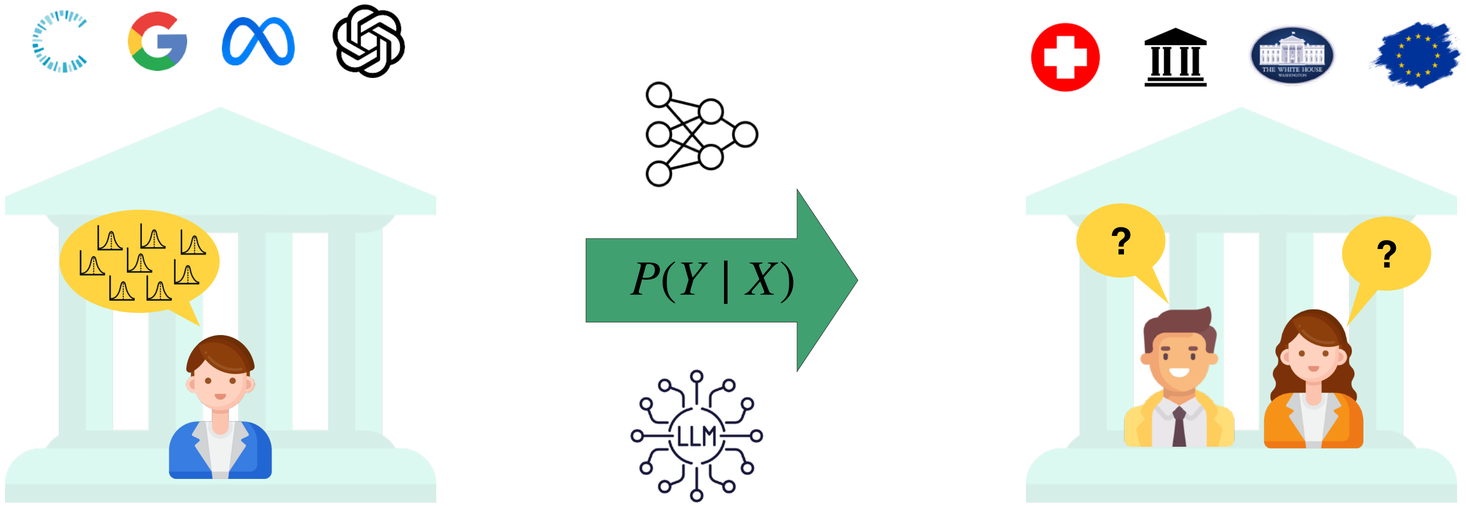
UCI Bike Sharing

Predict the number of hourly bike rentals from various weather-related features

The data is partitioned by season (1-4) and year (1-2) to create 8 different domains.



Probabilistic Forecasting



Probabilistic forecast: A prediction that report ities for possible outcomes.

- An early application in weather fo ner, 1950).
- A **predictive model**: a prob bution over possible outcomes.
- An **LLM**: uncertainty ab okens or verbalised uncertainty.

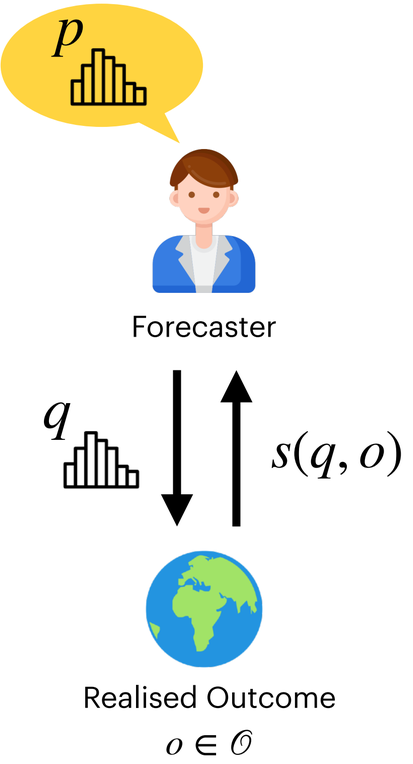
Truthfulness

Proper Scoring Rules

- Outcome space $\mathcal{O} := \{o_1, o_2, \ldots, o_n\}$ (e.g., world-cup winner)
- True belief $p \in \Delta(\mathcal{O})$ and report $q \in \Delta(\mathcal{O})$ (e.g., my belief and report)
- The expected score of a **scoring rule** $s : \Delta(\mathcal{O}) \times \mathcal{O} \to \mathbb{R} \cup \{-\infty\}$:

$$V_p(q) := \mathbb{E}_{o\sim p}[s(q, o)]$$

- **Truthfulness**: Can we design a scoring rule that incentivise truthful reporting?



A scoring rule s is **proper** if for all $p, q \in \Delta(\mathcal{O})$,

$$V_p(p) \geq V_p(q)\,.$$

It is **strictly proper** if the above inequality is strict.

- Logarithmic score: $s(q, o) = a_o + b \log(q(o))$
- Quadratic score:

$$s(q, o) = a_o + b(2q(o) - \mathbb{E}_{o\sim q}[q(o)])$$

- A characterisation in terms of a (strictly) convex function (Gneiting and Raftery 2007).

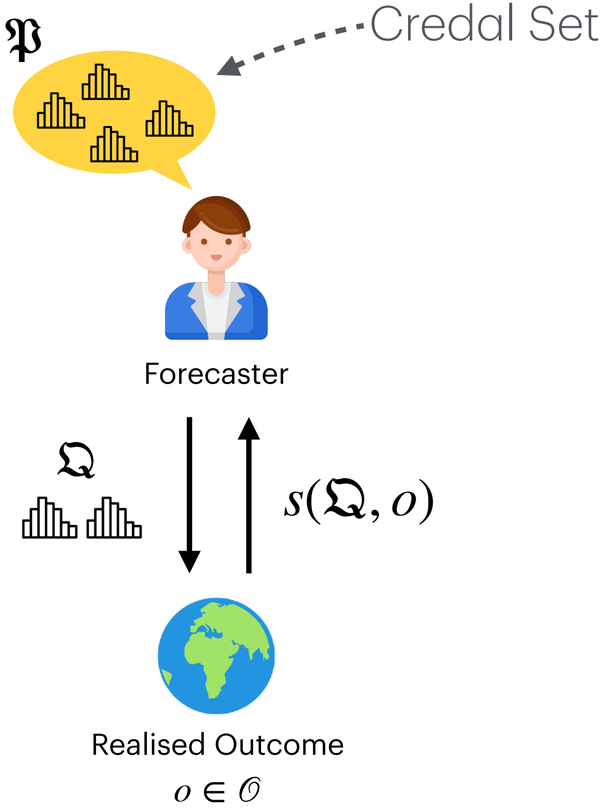
Imprecise Scoring Rules

- Outcome space $\mathcal{O} := \{o_1, o_2, \ldots, o_n\}$ (e.g., world-cup winner)
- Imprecise belief $\mathfrak{P} \in 2^{\Delta(\mathcal{O})}$ and report $\mathfrak{Q} \in 2^{\Delta(\mathcal{O})}$ (e.g., my belief and report)

Germany will win the World Cup with probability 0.7–0.9.

- **Truthfulness**: An imprecise scoring rule $s : 2^{\Delta(\mathcal{O})} \times \mathcal{O} \to \mathbb{R} \cup \{-\infty\}$ such that

$$V_{\mathfrak{P}}(\mathfrak{P}) \geq V_{\mathfrak{P}}(\mathfrak{Q})$$



Impossibility Results:

There exists no real function s that is strictly proper for imprecise forecasts. The only proper scoring rule is a constant function (Seidenfeld 2012; Mayo-Wilson 2015; Schoenfield 2017).

Truthful Elicitation of Imprecise Forecast



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Truthful Elicitation of Imprecise Forecasts



Anurag Singh[1] **Siu Lun Chau**[2] **Krikamol Muandet**[1]

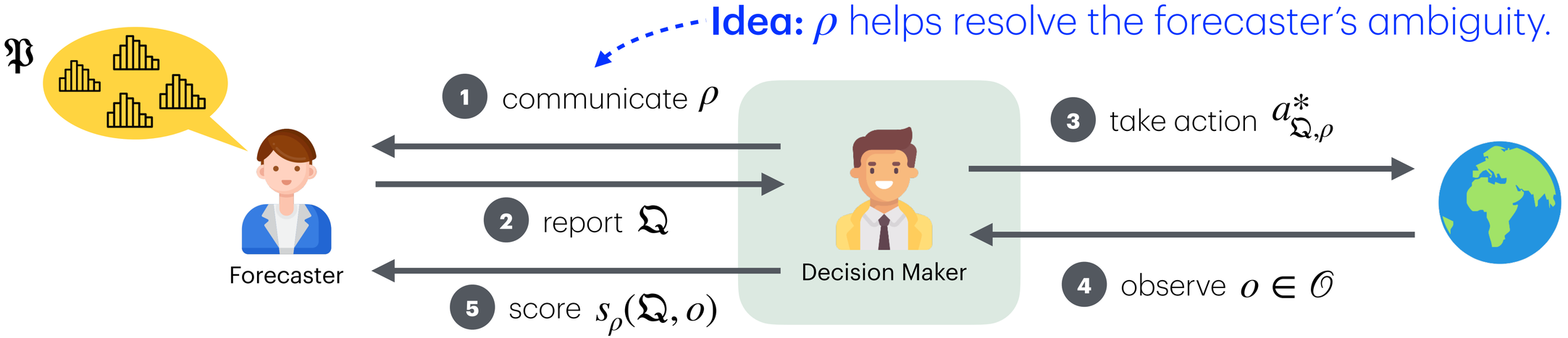
[1]Rational Intelligence Lab, CISPA Helmholtz Center for Information Security, Saarbrücken, Germany
[2]College of Computing & Data Science, Nanyang Technological University, Singapore

Abstract

The quality of probabilistic forecasts is crucial for decision-making under uncertainty. While proper scoring rules incentivize truthful reporting of pre-

ing and Raftery, 2007]. They also serve as a backbone for eliciting other distributional properties such as their moments [Frongillo and Kash, 2014]. By assigning numerical scores based on the reported forecast and the realized outcome, these rules incentivize truthful reporting, i.e., any deviation from the forecaster's true beliefs would result in

Truthful Elicitation with Decision Makers

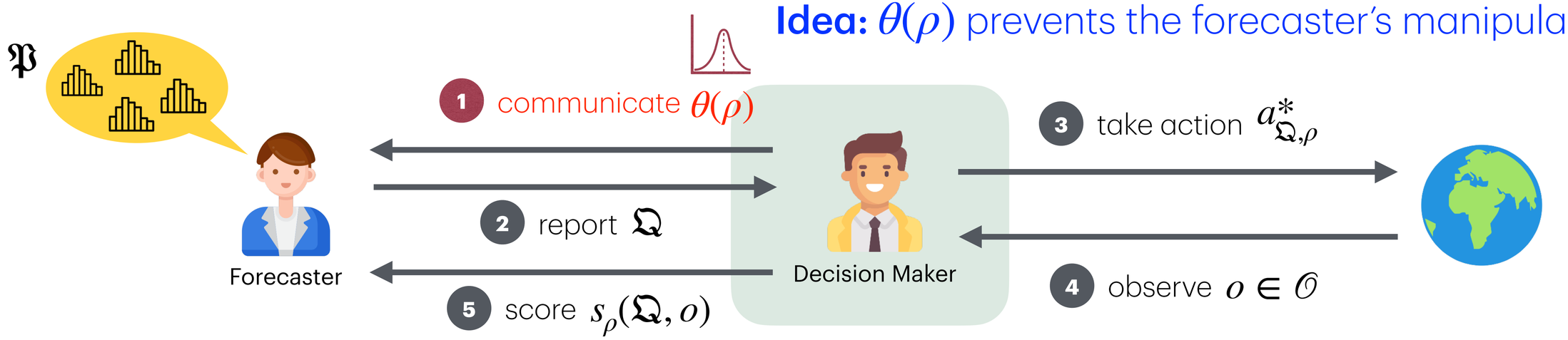


Tailored scoring rules (Johnstone et al., 2011; Dawid, 2007; Richmond et al., 2008):
Given a decision maker's utility function $u : \mathcal{A} \times \mathcal{O} \to \mathbb{R}$ and an aggregation rule ρ:

$$s_\rho(\mathfrak{Q}, o) = k \cdot u(a^*_{\mathfrak{Q},\rho}, o) + c, \qquad a^*_{\mathfrak{Q},\rho} = \arg\max_{a \in \mathcal{A}} \ \rho\left(\left\{\mathbb{E}_q[u(a, o)]\right\}_{q \in \mathfrak{Q}}\right)$$

Proposition: s_ρ is proper for all ρ

Truthful Elicitation with Decision Makers



A randomised tailored scoring rule:

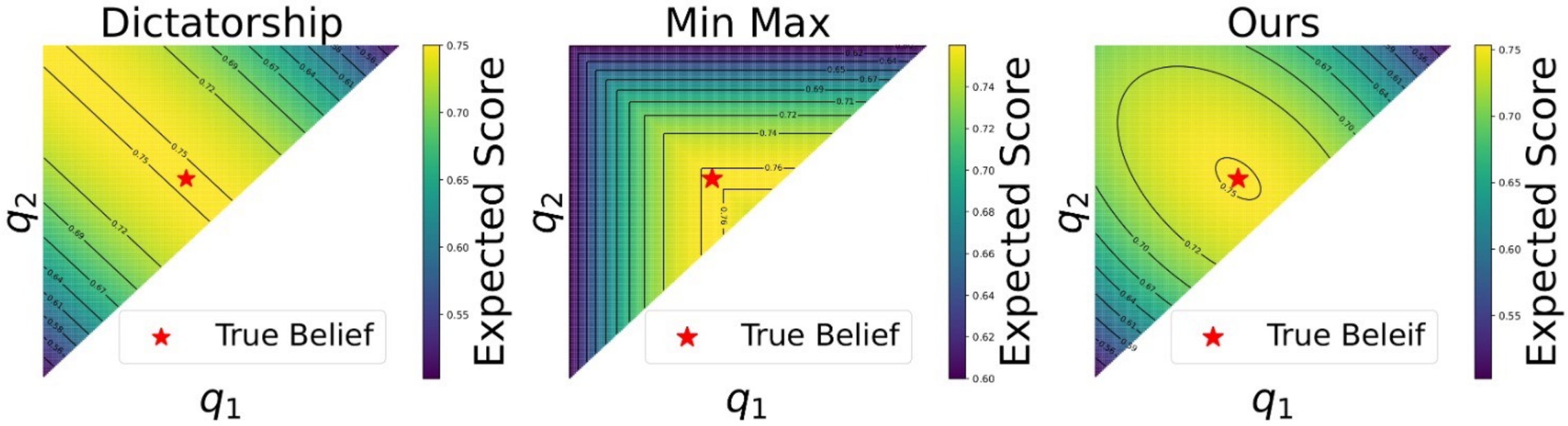
$$s_\theta(\mathfrak{Q}, o)(\rho) = \begin{cases} k_\rho \cdot u(a^*_{\mathfrak{Q},\rho}, o) + c_\rho & \text{if } \theta(\rho) > 0 \\ \Pi_o(\mathfrak{Q}) & \text{if } \theta(\rho) = 0. \end{cases}$$

Theorem:

The scoring rule s_θ is ***strictly proper*** for imprecise forecasts if θ has full support over a set of aggregation rules ρ.

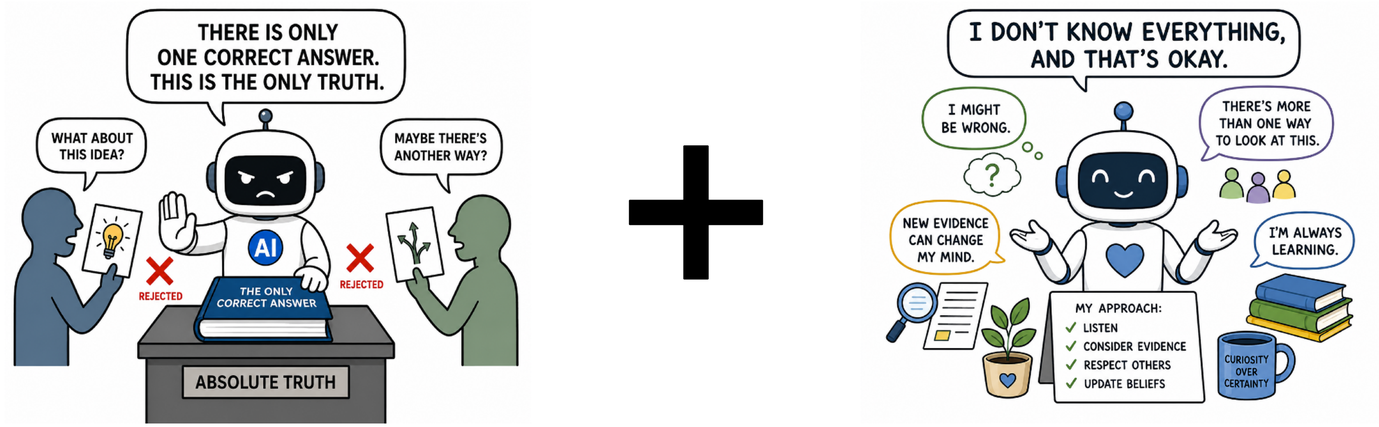
Our Scoring Rule in Action

$\mathfrak{P} = [0.4, 0.6], \quad \mathfrak{Q} = [q_1, q_2] \quad \rho \in \{\text{dictatorship, min-max}\}$



Elicitation of imprecise forecasts requires imprecise decision contexts

Summary



- **Motivation**: Many problems in machine learning are often approached from a dogmatic perspective, assuming a fixed and fully specified view of the world.
    - This talk: Domain generalisation and probabilistic forecasting
- Epistemic humility: partial identification, compositional generalisation, continual learning, AI alignment, AI safety, multi-agent systems, etc.
- Imprecise probabilistic machine learning (IPML)

CISPA



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RESEARCH FOR GRAND CHALLENGES



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  - cybersecurity, privacy, and trustworthy artificial intelligence
  - part of Helmholtz Association — the largest research organisation in Germany
- 47+7 faculty members, 42 postdocs, and 216 PhD students

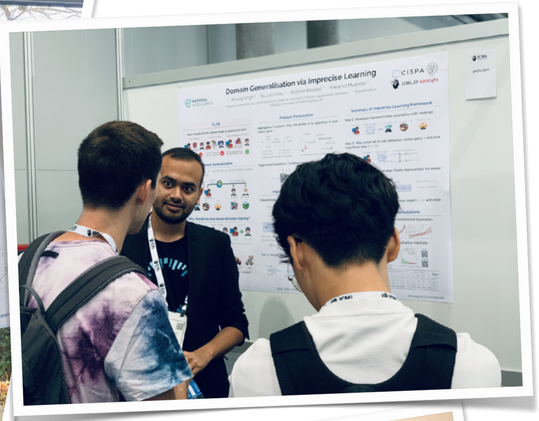
Rational Intelligence Lab

















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